



DAV — Voice-Assisted Design

User Manual

DAV is an inclusive development designed to promote the participation of people with motor disabilities in 2D/3D design.

The goal of the project is to enable people with motor disabilities to create and modify 3D models, drawings, and parts through spoken instructions. In doing so, it seeks to offer a voice-based alternative that complements the keyboard and mouse within the CAD environment, broadening opportunities for participation and promoting technological accessibility.



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User Manual – DAV Project

Introduction

The DAV (Voice-Assisted Design) project is a tool designed to facilitate interaction with CAD systems through voice commands, providing a more accessible and integrated design experience.

Getting Started

The system is based on FreeCAD 1.1.1. Make sure this version is installed to ensure compatibility with our voice control layer.

Interface and Commands

Graphical Interface: Developed with PySide, it allows users to view the object tree and configure preferences.

Voice Control: Uses the Vosk engine. The system recognizes specific commands that are translated into actions within FreeCAD.

Display Modes: Supports both light and dark modes, integrated with the software's native settings.

Main Features

- 2D and 3D Drawing (Workbenches: Draft, Part, Part Design, Sketcher).
- Voice navigation between files and printing.
- Voice-based object selection with visual feedback in the object tree.
- Multilingual support (Spanish, English, Portuguese).

Assistance and Preferences

Use the preferences panel to adjust the speech model based on your system's available RAM (small models for low consumption, large for higher accuracy).

General Operation of Voice Navigation

The DAV system operates through **hierarchical contexts** (level navigation):

1. **Initial access:** From the Base level, say «**explorer**» or «**file**» to enter the Explorer submenu, or «**selection**» to access object selection control.
2. **Submenus (enter):** When pronouncing a category (e.g. «*file*», «*edit*»), the system descends to that level and exclusively enables its commands.
3. **Executable commands:** Perform the action in FreeCAD when spoken.
4. **Go up a level:** In any subcontext, saying «**back**», «**go back**» or «**up**» returns to the upper level.
5. **View status:** Saying «**context**» or «**where am I**» lists the available options on screen.

Explorer Dictionary

General context icon: Dav/dic/Explorer/explorer.svg

Direct Commands (at Explorer root)

These commands are executed directly while staying in `explorer`, without needing to enter subfolders.

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	refresh , reload , update	Recalculates and updates the object tree and 3D view of the active document.	Open document with pending recalculation changes.
	screenshot , photo , take photo , screen capture , save screen	Opens the dialog box to save an image of the current 3D view.	An open document with active 3D view.
	text document , document , text	Creates an integrated text note/document object inside the project tree.	Active document.
	unlink , remove link , detach	Breaks the link of the selected object with respect to its original source.	Have a Link-type object selected.
	freeze , immobilize , lock	Immobilizes the selected object to prevent its automatic recalculation.	Selected object in the tree.
	all instances , select instances , all instances	Selects all shared instances of the current object in the view tree.	Object with multiple instances in the scene.
	variable set , variables , var set	Creates a special VarSet container to store global expressions and variables.	Active document.

File Submenu (File)

Voice access: file , files , folder , folders , sheets

Submenu icon: Dav/dic/Explorer/file.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	new , new file , create , create file	Creates a new blank project/document.	None.
	open , open file	Opens the system file explorer window to load a project.	None.
	save , save file	Saves changes of the active document to its previous path.	Active document with assigned path.

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	save as , save document as	Opens the dialog box to save the active document with a new name or path.	Active document.
	save copy , duplicate file	Saves a copy of the document without changing the current working file.	Active document.
	revert , restore	Discards unsaved changes and restores the last version saved to disk.	Previously saved active document.
	merge , join projects	Incorporates the content of another FCStd file into the current document.	Active document.
	import , import file	Imports external 2D/3D models (STEP, IGES, DXF, OBJ, etc.).	Active document.
	export , export file	Exports selected objects to external 2D/3D formats.	Selected object in the document.
	recent , recent files	Displays the floating menu with the last opened documents.	None.
	load image , open image	Allows inserting a bitmap image into the 3D work plane as a reference plane.	Active document.

Edit Submenu (Edit)

Voice access: edit , editing , modify , alter

Submenu icon: Dav/dic/Explorer/edit.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	undo , revert , go back	Reverts the last action performed in the project.	Action history available.
	redo , advance	Reapplies the last undone action.	Undo history active.
	cut	Copies to the clipboard and removes the selected object.	Selected object in the tree.
	copy	Copies current selection to the clipboard.	Selected object.
	paste	Pastes clipboard content into the active document.	Previous element in clipboard.

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	duplicate , clone	Creates an identical and independent copy of selected objects.	Selected object.
	select all , capture all	Selects all objects in the active document.	Document with objects.
	delete , remove , erase	Deletes selected objects or elements from the document.	Selected object.
	placement , position , set position	Opens the dialog box to enter numerical position and rotation coordinates.	Selected 3D object.
	transform , move	Activates visual 3D gimbals/manipulators to move and rotate the object interactively.	Selected 3D object.
	align , alignment	Opens the tool to geometrically align two or more objects.	Two or more selected objects.
	preferences , settings	Opens general program preferences panel.	None.
	properties , details	Focuses the property panel (Data/View) of the selected object.	Selected object.
	send to python , python console	Sends selected object reference to Python command console.	Selected object.
	edit mode , modify mode	Enters specialized interactive object editing mode.	Editable object (sketch, mesh, etc.).

Print Submenu (Print)

Voice access: print , printing , printer , pdf , generate pdf , export pdf

Submenu icon: Dav/dic/Explorer/printcmds.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	print , printer	Opens the physical printing window for the current view or technical sheet.	Active 2D/3D view or TechDraw sheet.

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	pdf , export pdf , generate pdf	Exports the current view or technical sheet directly into a vectorized PDF document.	Active document or view.

Windows Submenu (Windows)

Voice access: windows , window

Submenu icon: Dav/dic/Explorer/Windows/close.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	close , close tab , close document	Closes the currently active tab or document.	At least one open document.
	close all , close all tabs	Consecutively closes all open project tabs.	Open documents.
	exit , close freecad , close application	Completely closes the FreeCAD application after save confirmation.	None.

Expressions Submenu (Expressions)

Voice access: expressions , expression

Submenu icon: Dav/dic/Explorer/expression.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	copy document , copy document name	Copies active document name reference for use in parameter formulas.	Saved document.
	copy all , copy all expressions	Copies all expression references available in the tree.	Configured expressions.
	copy selection, copy selected expression	Copies the parameter formula path of the selected object.	Selected parameter or object.
	paste expression , apply formula	Pastes and binds saved expression to the selected property.	Property opened in editing.

Tools Submenu (Tools)

Voice access: tools , tool , utilities

Submenu icon: Dav/dic/Explorer/Tools/customize.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	measure , measurement , measure distance	Activates interactive virtual ruler to measure distances and angles in 3D space.	3D elements in view.
	clear selection , clean selection	Pops up a menu categorizing selected elements by geometry type.	Complex/multiple selection.
	demo mode , turn model , continuous rotation	Starts a continuous panoramic camera rotation of the 3D model for presentation.	Object in 3D view.
	customize , customization	Opens dialog box to modify keyboard shortcuts, toolbars, and user commands.	None.
	edit parameters , parameter editor	Opens advanced Parameter Editor for FreeCAD configuration system.	None.
	project utilities , project utility	Opens utility to decompress and repair internal XML structure of .FCStd files.	None.

Structure Submenu (StructureToolbar)

Voice access: structure , structure bar

Submenu icon: Dav/dic/Explorer/structure.svg

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	part , create part , part	Creates a Part structural container to group multiple bodies or solids.	Active document.
	group , new group , create group	Creates a generic group folder in tree without special geometric properties.	Active document.
	link actions , link , link submenu	Opens a submenu dedicated to creating and managing linked instances (Links).	Object in document.

Secondary Submenu linkActions (inside Structure):

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	create link , make link	Generates a lightweight link (App::Link) to an existing object.	Selected object.

SVG Icon	Voice Commands (Synonyms)	What does it do?	Requirements / Prerequisites
	relative link , link relative	Generates a relative sub-element link.	Selected object/sub-element.
	link group , links group	Groups multiple links in a group container.	Active document.
	import link , bring link	Imports an external object as a link.	Active document.
	import all links , import links	Imports all linked elements into the active file.	External links in the project.
	recalculate link , update link	Forces updating of the selected link.	Selected link.

Selection Tool (selection)

The **Selection** tool is a voice-accessible navigation module implementing the `ObjectSelection` class. It allows sequentially traversing all active document objects via keyboard or microphone without using the mouse pointer in the tree view.

How the objectSelection Engine Works

1. Maintains an ordered list of all loaded objects in the active project tree.
2. When invoking voice commands, an internal pointer moves forward or backward element by element.
3. The reached object is automatically highlighted in the 3D view and selected in the model tree.

Selection Tool Voice Commands

Voice access from Base: selection , navigate objects

SVG Icon	Voice Commands (Synonyms)	Engine Action	What does it do?	Requirements / Prerequisites
-	next , advance , another , pass , next object , object next , next element , select next	SelectNext()	Advances to the next object in the document list and selects it on screen.	At least one object in the document.

SVG Icon	Voice Commands (Synonyms)	Engine Action	What does it do?	Requirements / Prerequisites
-	previous , go back , return , previous object , previous object , previous element , select previous	SelectPrevious()	Steps back to the previous object in the list and selects it on screen.	At least one object in the document.
	all , everything , select all , select all , select all objects	SelectAll()	Simultaneously selects all objects present in the document (shares Edit selectall.svg icon).	Document with objects.
-	nothing , none , clear , clear all , unmark , unmark all , deselect , clear selection	DeselectAll()	Clears any current selection, leaving scene with no objects selected.	None.
-	this , current , current object , which , what object do I have	CurrentObject()	Displays via console or speech synthesizer the name of the object where pointer is focused.	Have initialized object list.
-	how many , count , how many objects , how many are there	ObjectCount()	Reports total quantity of registered objects in active document.	Active document.
-	help	ayuda()	Displays quick help for selection commands in console.	None.

Summary of File Locations in Repository

Explorer Dictionary: Dav/dic/Explorer/Explorer.py and Dav/dic/Explorer/TraduceToEs.py

Explorer SVG Icons: Located in Dav/dic/Explorer/ and corresponding subfolders (File/ , Edit/ , Print/ , Windows/ , Expressions/ , Tools/ , StructureToolbar/).

Selection Dictionary: Dav/dic/Selection/selection.py and Dav/dic/Selection/TraduceToES.py

Selection Backend Engine: Dav/scr/selection/object_selection.py

Global Navigation Commands: Dav/dic/NavCommands/TraduceToEs.py

Complete FreeCAD Command Guide (DAV Project)

Technical tables structured by workbench, technical views, drawings, symbology, dimensions, solids, and sketching.

1. General Commands / Base Level

Global access available at all times within the navigation tree.

Icon	Command	Description	Requirements
	explorer / file	Accesses the file management, editing, and window submenu.	None.
	views / stdview	Opens the submenu for standard views, camera controls, and appearance.	Active document.
	workbench / selector	Opens the workbench selector (PartDesign, Draft, Sketcher, Part, Assembly, TechDraw).	None.
	line attributes / lineattributes	Accesses configuration for line style, thickness, and color.	None.
	selection	Enables the sequential navigation engine and voice-driven object selection.	Active document containing objects.
	preferences	Opens the general configuration and preferences window for the DAV system.	None.

2. Assembly Workbench

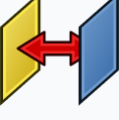
Core commands for creating and managing mechanical assemblies.

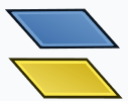
Ícono	Comando (Español)	¿Qué hace?	Requisitos
	create assembly / create	Creates a new Assembly container in the active document.	Active document.
	insert part / newpart	Creates and inserts a new Part within the active assembly.	Active assembly container.
	insert link / link	Inserts an existing component using a lightweight Link into the assembly.	Active assembly and existing components.
	solve	Runs the kinematic solver to calculate and update constraints.	Assembly with applied joints.
	create view / view	Generates an exploded view or custom view configuration of the assembly.	Active assembly.

Ícono	Comando (Español)	¿Qué hace?	Requisitos
	create simulation / simulation	Creates a kinematic motion simulation container.	Assembly with configured joints.
	bill of materials / bom	Generates the Bill of Materials (BOM) table for the assembly.	Assembly with parts.
	ground / grounded	Fixes and locks the selected part to act as a stationary base.	Selected component in the assembly.
	assembly preferences / preferences	Opens the preferences panel for the Assembly module.	None.
	assembly help / help	Displays contextual help for the Assembly workbench.	None.

3. Assembly Joints

Kinematic constraints and mechanical couplings between components.

Ícono	Comando (Español)	¿Qué hace?	Requisitos
	fixed joint / fixed	Locks all relative degrees of freedom between two selected elements.	Two selected parts, faces, or edges.
	revolute joint / revolute	Allows rotation exclusively around a shared axis.	Two selected circular/cylindrical geometries.
	slider joint / slider	Enables linear displacement along a rectilinear axis or guide.	Two selected linear edges/axes.
	cylindrical joint / cylindrical	Allows combined rotation and translation along the same axial axis.	Two selected cylindrical surfaces.
	ball joint / ball	Enables multidirectional spherical rotation while preserving a fixed center.	Two selected spherical surfaces or points.
	distance joint / distance	Defines and constrains a fixed specific distance between two geometries.	Two selected geometric entities.

Ícono	Comando (Español)	¿Qué hace?	Requisitos
	angle joint / angle	Fixes a specific angle between two planes, faces, or edges.	Two selected planar faces or edges.
	parallel joint / parallel	Constrains two planes or edges to remain perpetually parallel.	Two selected planar faces or edges.
	perpendicular joint / perpendicular	Constrains two elements to maintain a 90° right angle.	Two selected planar faces or edges.
	gears joint / gears	Kinematically couples the rotation of two gear wheels or shafts.	Two pre-existing revolute joints.
	belt joint / belt	Couples rotational transmission between two pulleys or shafts.	Two revolute joints in the assembly.
	rack and pinion / rackpinion	Couples the rotation of a pinion with the linear displacement of a rack.	One revolute joint and one slider joint.
	screw joint / screw	Couples linear travel with helical rotation according to thread pitch.	One cylindrical joint or helical pair.
	joints help / help	Displays contextual help for mechanical joints.	None.

4. TechDraw Workbench – Pages and Outputs (Page)

Canvas creation, standardized templates, redrawing, and exporting (DXF, SVG).

Icon	Command	Description	Requirements
	default page / default (page)	Inserts a new technical drawing page using the default standard template.	Active document.
	template page / template (page)	Opens the file browser to create a page with a custom SVG template.	Active document.
	redraw page / redraw (page)	Forces a full refresh and regeneration of all views on the technical sheet.	Active TechDraw page.

Icon	Command	Description	Requirements
	print page / print (page)	Opens the physical print dialog for all technical pages in the document.	At least one TechDraw page.
	export to dxf / dxf (page)	Exports the active technical drawing page to 2D vector CAD DXF format.	Active TechDraw page.
	export to svg / svg (page)	Exports the active technical sheet to an SVG vector graphic file.	Active TechDraw page.

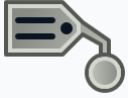
5. TechDraw Workbench – Projection and Section Views

Primary views, detail views, broken views, clip groups, complex sections, spreadsheets, and active 3D view.

Icon	Command	Description	Requirements
	insert view / view (views)	Inserts the standard orthogonal 2D projection of the selected solid or object.	Solid/object and active page.
	detail view / detailview (views)	Inserts a magnified circular or rectangular detail view of a region.	Selected existing drawing view.
	broken view / brokenview (views)	Creates a continuity break/interrupted view for long or slender parts.	Selected existing drawing view.
	clip group / clipgroup (views)	Creates a clipping group to frame and limit the visible boundary of views.	Active TechDraw page.
	complex section / complexsection (views)	Generates a cross-section view based on a stepped profile or 2D sketch.	Selected 3D object and cutting sketch.
	sketch view / draft (views)	Inserts a direct 2D projection of a Sketcher sketch or Draft wire onto the page.	Selected sketch/Draft object.
	spreadsheet view / spreadsheet (views)	Inserts a rendered table linked to a spreadsheet object.	Spreadsheet in active document.
	active 3D view / active_view (other_views)	Inserts a bitmap screenshot of the current 3D camera orientation into the page.	Active 3D view and TechDraw page.

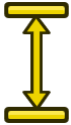
6. TechDraw Workbench – Symbolology, Text, and Hatching

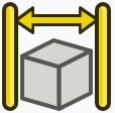
Standardized welding symbols, surface finishes, rich text, and geometric hatching.

Icon	Command	Description	Requirements
	welding symbol / weldsymbol (symbols)	Inserts a standard engineering welding symbol with leader line and parameters.	Technical view and selected edge/point.
	rich text / richtext (symbols)	Inserts a formatted text block (HTML/Markdown, bold, lists) into the sheet.	Active TechDraw page.
	surface finish / finish (symbols)	Inserts standardized surface roughness and finish symbols (machining, knurling).	Selected edge or view.
	geometric hatch / geometric_hatch (hatching)	Applies a standardized geometric hatching/shading pattern to a closed face.	Closed face in TechDraw view.

7. TechDraw Workbench – Dimensions

2D dimensioning commands on technical drawing views (length, angles, radii, diameters, tolerances).

Icon	Command	Description	Requirements
	length dimension / dimension / length	Inserts an aligned distance/length dimension on an edge in the drawing view.	Selected edge in TechDraw view and active page.
	horizontal dimension / horizontal (techdraw)	Inserts a horizontal distance dimension (X) between two vertices or an edge.	Selected edge or two vertices in TechDraw view.
	vertical dimension / vertical (techdraw)	Inserts a vertical distance dimension (Y) between elements in the sheet.	Selected elements in TechDraw view.
	angle dimension / angle (techdraw)	Inserts an angular dimension defined between two selected edges.	Two selected edges in TechDraw view.
	3-point angle dimension / points (angle techdraw)	Inserts an angular dimension defined sequentially across three vertices.	Three selected vertices in TechDraw view.
	diameter dimension / diameter (techdraw)	Inserts a diameter dimension onto a circular or cylindrical edge.	One circular edge selected in TechDraw view.

Icon	Command	Description	Requirements
	radius dimension / radius (techdraw)	Inserts a radius dimension onto an arc or circular boundary.	One circular edge/arc selected in TechDraw view.
	extent dimension / extent (techdraw)	Inserts a maximum extent / overall boundary dimension between two edges.	Two selected edges in TechDraw view.
	area dimension / area (techdraw)	Calculates and inserts an area annotation for the selected face.	Selected planar face in TechDraw view.
	hole/shaft fit / fit (techdraw)	Inserts a standardized engineering fit and tolerance symbol (Hole/Shaft Fit).	Selected dimension or hole/shaft edge.

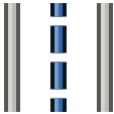
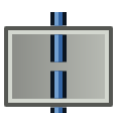
8. TechDraw Workbench – Annotations, Balloons, and Templates

Insertion of text notes, axonometric dimensions, BOM callout balloons, symbols, images, and title block fields.

Icon	Command	Description	Requirements
	text annotation / annotation (techdraw)	Inserts a free text block or informative note onto the technical page.	Active TechDraw page.
	axonometric dimension / axo_length (techdraw)	Inserts a projected length dimension tailored for axonometric/isometric views.	Selected edge in an isometric view.
	balloon / balloon (techdraw)	Inserts a callout balloon with reference number/item mark for Bill of Materials.	Selected element or point in the view.
	fill template fields / fields (features)	Opens the editor to fill editable text fields in the page's title block.	Active TechDraw page with template.
	insert image / image (features)	Inserts a bitmap graphic image (logo, diagram) into the technical sheet.	Active TechDraw page.
	insert symbol / symbol (features)	Inserts a standard SVG vector symbol onto the technical page.	Active TechDraw page.

9. TechDraw Workbench – Centerlines, Reference Points, and Topology

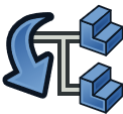
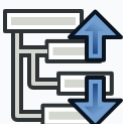
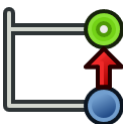
Centerlines, cosmetic lines, vertices, midpoints, quadrants, and topological visibility.

Icon	Command	Description	Requirements
	centerline between two lines / twolines (addlines)	Inserts a centerline or symmetric axis between two selected lines.	Two selected lines or edges in TechDraw view.
	centerline between two points / twopoints (addlines)	Inserts a centerline between two selected points or vertices.	Two selected vertices in TechDraw view.
	cosmetic line / cosmetic (addlines)	Draws an auxiliary 2D cosmetic line between two vertices.	Two selected vertices on the technical sheet.
	decorate line / decorate (addlines)	Modifies the color, weight, or style pattern of the selected line.	Selected line or edge in TechDraw.
	face center lines / center (addlines)	Generates primary centerlines across a cylindrical or circular face.	Selected face in the drawing view.
	cosmetic vertex / cosmetic (addvertices)	Inserts a cosmetic reference point or vertex on the technical page.	Active TechDraw view.
	midpoints / midpoints (snaps)	Toggles and displays midpoint snap reference points on view edges.	Selected TechDraw view.
	quadrants / quadrants (snaps)	Toggles and displays the 4 quadrant cardinal points on circles and arcs.	Selected TechDraw view.
	show all / showAll (topology)	Toggles visibility for all hidden edges, seams, and invisible lines.	Selected TechDraw view.

10. PartDesign Workbench – Base Structure and Management

Bodies creation, shape binders, feature management, and design tree.

Icon	Command	Description	Requirements
	body / create body	Creates an active Body container (PartDesign Body) for solid part modeling.	Active document.

Icon	Command	Description	Requirements
	new sketch / newsketch	Creates a new sketch directly attached to the active body or support plane.	Active body or selected plane.
	clone	Creates a linked parametric clone of a PartDesign body or element.	Selected body.
	sub-shape binder / subshapebinder	Creates a linked parametric reference (SubShapeBinder) of external geometry.	Selected external geometry.
	move feature / movefeature	Moves a design feature to another body or tree position.	Selected feature.
	move feature in tree / movefeatureintree	Reorders the hierarchical position of a feature within the dependency tree.	Selected feature in tree.
	set tip / movetip	Sets the active calculation endpoint (Tip) of the body to the selected feature.	Selected feature within the body.
	shaft wizard / wizardshaft	Opens the interactive wizard for stepped shaft design and calculation.	Active body.
	design preferences / preferences	Opens the PartDesign / System preferences panel.	None.

11. PartDesign Workbench – Additive Operations and Primitives

Adding material onto active bodies via profile extrusion or solid primitives.

Icon	Command	Description	Requirements
	pad / extrude pad	Adds material by extruding a closed sketch or face along a normal direction.	Closed sketch or planar face in active body.
	additive revolution / revolution	Adds material by revolving a closed sketch around an axis of revolution.	Closed sketch and revolution axis in active body.
	additive helix / additivehelix	Generates an additive thread or spring by sweeping a profile along a helix.	Profile sketch and helical axis in active body.

Icon	Command	Description	Requirements
	additive loft / additiveloft	Creates an additive solid by smooth transition across multiple cross-sections.	Minimum 2 section sketches in active body.
	additive pipe / sweep / additivepipe	Adds material by sweeping one or more profiles along a guide path.	Profile and guide path in active body.
	additive box / additivebox	Inserts a fused additive box/prism primitive into the active body.	Active body (or creates a new body).
	additive cone / additivecone	Inserts an additive cone or truncated cone primitive into the active body.	Active body (or creates a new body).
	additive cylinder / additivecylinder	Inserts an additive fused cylinder primitive into the active body.	Active body (or creates a new body).
	additive ellipsoid / additiveellipsoid	Inserts an additive parametric ellipsoid primitive into the active body.	Active body (or creates a new body).
	additive prism / additiveprism	Inserts an additive parametric polygonal prism primitive into the active body.	Active body (or creates a new body).
	additive sphere / additivesphere	Inserts an additive parametric sphere primitive into the active body.	Active body (or creates a new body).
	additive torus / additivetorus	Inserts an additive parametric torus primitive into the active body.	Active body (or creates a new body).
	additive wedge / additivewedge	Inserts an additive pyramidal wedge primitive into the active body.	Active body (or creates a new body).

12. PartDesign Workbench – Subtractive Operations and Primitives

Cutting, pocketing, and removing material via negative profiles or solid primitives.

Icon	Command	Description	Requirements
	pocket	Removes material by extruding a sketch inward into the solid.	Closed sketch on planar face or active body.

Icon	Command	Description	Requirements
	groove	Removes material by revolving a sketch around an axis of revolution.	Closed sketch and axis in active body.
	hole	Creates standardized simple, threaded, countersunk, or counterbored holes.	Sketch with point centers or circular edges.
	subtractive box / subtractivebox	Removes material by subtracting a box primitive from the solid.	Existing active solid/body.
	subtractive cone / subtractivecone	Removes material by subtracting a cone primitive from the solid.	Existing active solid/body.
	subtractive cylinder / subtractivecylinder	Removes material by subtracting a cylindrical shape from the solid.	Existing active solid/body.
	subtractive ellipsoid / subtractiveellipsoid	Removes material by subtracting an ellipsoid primitive from the solid.	Existing active solid/body.
	subtractive helix / subtractivehelix	Removes material by sweeping a profile along a helix (internal threads).	Profile and helical axis in active body.
	subtractive loft / subtractiveloft	Removes material by smooth transition across multiple cutting profiles.	Minimum 2 profiles in active body.
	subtractive pipe / subtractivepipe	Removes material by sweeping a profile along a cutting trajectory.	Profile and trajectory in active body.
	subtractive prism / subtractiveprism	Removes material by subtracting a polygonal prism primitive from the solid.	Existing active solid/body.
	subtractive sphere / subtractivesphere	Removes material by subtracting a sphere primitive from the solid.	Existing active solid/body.
	subtractive torus / subtractivetorus	Removes material by subtracting a torus primitive from the solid.	Existing active solid/body.

Icon	Command	Description	Requirements
	subtractive wedge / subtractivewedge	Removes material by subtracting a wedge primitive from the solid.	Existing active solid/body.
	design boolean / boolean	Performs Boolean operations (Union, Cut, Intersection) between design bodies.	Two bodies in the active document.

13. PartDesign Workbench – Modification and Transformation

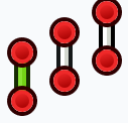
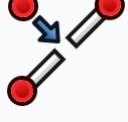
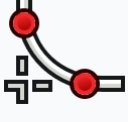
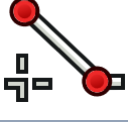
Dress-up features (fillets, chamfers, drafts, thickness) and repetitive patterns.

Icon	Command	Description	Requirements
	fillet / fillet feature	Applies a fillet or rounded blend radius to selected edges of the body.	Selected edges in active body.
	chamfer / chamfer feature	Applies a beveled edge or chamfer to selected edges of the body.	Selected edges in active body.
	draft / draft angle	Applies a draft/taper angle to selected planar faces for mold release.	Selected faces and neutral plane.
	thickness / hollow shell	Transforms the solid into a hollow shell by assigning a uniform wall thickness.	Faces to open selected in active body.
	linear pattern / linearpattern	Repeats a feature along a linear direction with defined spacing.	Prior feature and directional edge/axis.
	mirrored / design symmetry	Generates a mirrored copy of features with respect to a reference plane.	Feature and symmetry plane in active body.
	polar pattern / polarpattern	Repeats a feature radially around a central rotational axis.	Feature and rotation axis in active body.
	multi-transform / multitransform / scaled	Applies compounded transformations and chained scaling to features.	Selected prior feature in active body.

14. Sketcher Workbench - Operations and Environment Control

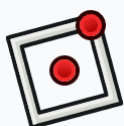
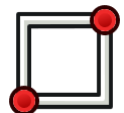
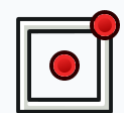
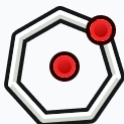
Sketching environment controls, element manipulation, edge editing, and clipboard.

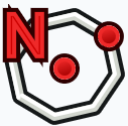
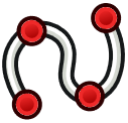
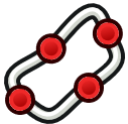
Icon	Command	Description	Requirements
	new sketch / new	Creates a new sketch on the selected 2D plane.	Active document.
	edit sketch / edit	Enters interactive edit mode for the selected sketch.	Selected sketch.
	attach / map / attach	Maps or attaches the sketch onto a planar face or reference plane.	Selected sketch and target plane.
	grid / toggle grid	Toggles visibility and grid-snapping within the sketch viewport.	Active sketch environment.
	stop operation / stop	Cancels or terminates the current interactive drawing tool.	Active drawing tool.
	leave sketch / leave	Closes sketch editing mode and triggers a model recalculation.	Active sketch environment.
	toggle construction / toggleconstruction	Toggles selected geometry between standard and auxiliary construction mode.	Selected geometries in sketch.
	cancel editing / cancelediting	Cancels current editing, discarding uncommitted modifications.	Active sketch environment.
	carbon copy / carboncopy	Copies all geometry and constraints completely from another sketch.	Active sketch environment.
	copy elements / copyelements	Copies selected sketch elements to the clipboard.	Selected sketch geometries.
	cut elements / cutelements	Cuts and copies selected sketch geometries to the clipboard.	Selected sketch geometries.
	paste elements / pasteelements	Pastes sketch geometries from the clipboard.	Active sketch environment.

Icon	Command	Description	Requirements
	mirror / symmetry	Creates a symmetrical copy of geometries across a sketch axis or line.	Selected geometries and symmetry axis.
	mirror sketch / mirrorsketch	Generates a duplicated mirrored sketch across a plane.	Selected sketch.
	offset	Creates an offset contour at an equidistant distance inside the sketch.	Selected sketch contour or edges.
	move array / movearray	Moves or generates multiple linear copies of selected entities.	Selected geometries.
	polar rotation / rotatepolar	Rotates or radially repeats sketch entities around a center point.	Selected sketch geometries.
	scale	Applies a dimensional scale factor to selected sketch geometries.	Selected geometries.
	trim edge / trimedge	Trims a line or curve segment to the nearest intersection.	Selected edge in sketch.
	split edge / splittedge	Splits a line or curve into two segments at the indicated point.	Selected edge in sketch.
	extend edge / extendedge	Extends a line or arc until reaching another geometric entity.	Selected edge in sketch.
	sketch fillet / fillet	Creates a tangential fillet or rounding between two intersecting lines.	Two lines or selected vertex in sketch.
	sketch chamfer / chamfer	Creates a straight beveled corner between two intersecting lines.	Two lines or selected vertex in sketch.

15. Sketcher Workbench – Geometric Entities (Geometry)

Tracing lines, polylines, rectangles, squares, triangles, slots, regular polygons, text, arcs, B-Splines, and conics.

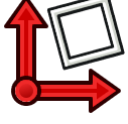
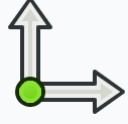
Icon	Command	Description	Requirements
	line / create (line)	Draws a straight 2D line segment between two points in the sketch.	Active sketch environment.
	line by points / create_by_points (line)	Creates a straight line segment by specifying parametric numeric coordinates.	Active sketch environment.
	polyline / create (polyline)	Draws a continuous sequence of connected line and arc segments (Polyline).	Active sketch environment.
	point / create (point)	Inserts a geometric reference point element into the sketch.	Active sketch environment.
	triangle / create (triangle)	Draws a regular equilateral triangle inside the sketch.	Active sketch environment.
	square / create (square)	Draws a regular four-sided square inside the sketch.	Active sketch environment.
	rectangle / create (rectangle)	Draws a rectangle defined by two opposing diagonal corners.	Active sketch environment.
	centered rectangle / center (rectangle)	Draws a rectangle defined from its center point and one corner.	Active sketch environment.
	slot / oblong / create (slot / oblong)	Draws a straight slot (Oblong) with semicircular rounded ends.	Active sketch environment.
	pentagon / pentagon (polygon)	Draws a regular 5-sided polygon (Pentagon) inscribed or circumscribed.	Active sketch environment.
	hexagon / create (hexagon)	Draws a regular 6-sided polygon (Hexagon).	Active sketch environment.
	heptagon / create (heptagon)	Draws a regular 7-sided polygon (Heptagon).	Active sketch environment.

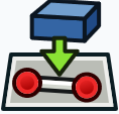
Icon	Command	Description	Requirements
	octagon / octagon (polygon)	Draws a regular 8-sided polygon (Octagon).	Active sketch environment.
	regular polygon / regular (polygon)	Draws a regular polygon by defining an arbitrary number of sides and radius.	Active sketch environment.
	shape text / create (text / shapestring)	Opens the ShapeString tool to insert accessible geometric text into the sketch.	Active document and font file.
	arc by center / center (arc)	Draws a circular arc by specifying center point and endpoints.	Active sketch environment.
	arc by 3 points / 3point (arc)	Draws a circular arc defined sequentially through three points.	Active sketch environment.
	arc slot (round ends) / arc_ends (arc_slot)	Draws a curved arc slot with rounded circular ends.	Active sketch environment.
	arc slot (flat ends) / flat_ends (arc_slot)	Draws a curved arc slot with flat or square ends.	Active sketch environment.
	B-spline curve / create (bspline)	Creates a B-Spline curve defined via pole control points.	Active sketch environment.
	interpolated B-spline / interpolation	Creates a B-Spline curve that passes exactly through the indicated points.	Active sketch environment.
	periodic B-spline / periodic	Creates a closed and periodic B-Spline curve using control points.	Active sketch environment.
	interpolated periodic B-spline / periodicinterp	Creates a closed periodic B-Spline curve interpolated through given points.	Active sketch environment.
	convert to NURBS / tonurbs (bspline_tools)	Converts a B-Spline curve into a rational parametric NURBS spline.	Selected B-Spline curve in sketch.

Icon	Command	Description	Requirements
	decrease degree / decrease (bspline_tools)	Reduces the polynomial degree of the B-Spline curve by one.	Selected B-Spline curve in sketch.
	increase degree / increase (bspline_tools)	Increments the degree of the B-Spline curve by adding control poles.	Selected B-Spline curve in sketch.
	insert knot / knot (bspline_tools)	Inserts a new knot point into the parametric knot vector of the B-Spline.	Selected B-Spline curve in sketch.
	join curves / join (bspline_tools)	Joins two adjacent or continuous B-Spline curves into a single curve.	Two adjacent B-Spline curves.
	circle by center / create (circle)	Draws a circle defined by center point and radius point.	Active sketch environment.
	circle by 3 points / 3point (circle)	Draws a circle passing through three given points on the plane.	Active sketch environment.
	ellipse by center / center (ellipse)	Draws a full ellipse specifying center, major radius, and minor radius.	Active sketch environment.
	ellipse by 3 points / 3points (ellipse)	Draws an ellipse defined sequentially by three perimeter points.	Active sketch environment.
	elliptic arc / elliptic (ellipse)	Draws an elliptical arc segment specifying center, radii, and angles.	Active sketch environment.
	hyperbolic arc / hyperbolic (ellipse)	Draws a conic hyperbola curve or arc in the sketch.	Active sketch environment.
	parabolic arc / parabolic (ellipse)	Draws a conic parabola curve or arc in the sketch.	Active sketch environment.

16. Sketcher Workbench – Utilities, Validation, and Views

Geometry cleanup, orientation, validation, section views, axis selection, and external references.

Icon	Command	Description	Requirements
	delete constraints / deleteconstraints (tools)	Deletes all dimensional and geometric constraints from the active sketch.	Sketch containing constraints.
	delete geometry / deletegeometry (tools)	Deletes all geometric entities and elements from the active sketch.	Sketch containing geometry.
	merge sketches / merge (tools)	Combines two or more selected sketches into a single merged sketch.	Two or more selected sketches.
	reorient sketch / reorient (tools)	Opens dialog to reorient or reassign the support plane of the sketch.	Selected sketch.
	remove axes alignment / removeaxes (tools)	Removes automatic alignment locked to coordinate axes.	Active sketch.
	validate sketch / validate	Verifies sketch integrity, detects missing coincidences, and fixes tolerances.	Active sketch.
	view sketch / sketch (view)	Orients the 3D camera perpendicular to the sketch working plane.	Active sketch environment.
	section view / section (view)	Temporarily hides 3D solid geometry that obstructs the sketch plane.	Active sketch inside a 3D solid.
	origin / root / origin (root)	Selects the absolute origin point (0, 0) of the sketch.	Active sketch environment.
	select horizontal axis / horizontal (select)	Selects the horizontal X-axis of the sketch as a reference entity.	Active sketch environment.
	select vertical axis / vertical (select)	Selects the vertical Y-axis of the sketch as a reference entity.	Active sketch environment.
	select origin / origin (select)	Selects the Cartesian origin vertex of the sketching coordinate frame.	Active sketch environment.

Icon	Command	Description	Requirements
	external geometry / intersection (external)	Links and projects an external 3D edge or feature as an intersection reference.	Edge or face of external 3D solid.
	external projection / projection (external)	Projects external geometry directly onto the active sketch plane.	External solid feature or face.

17. Sketcher Workbench – Constraints

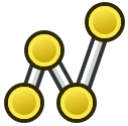
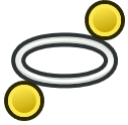
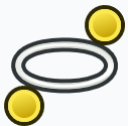
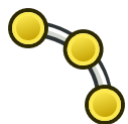
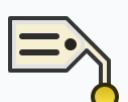
Dimensional constraints and fundamental geometric relationships.

Icon	Command	Description	Requirements
	distance dimension / dimension / distance	Applies a fixed linear distance constraint or general dimension to a line.	At least one line or two points selected.
	horizontal dimension / horizontal (dimensional)	Applies a horizontal length constraint restricted along the X-axis.	One line or two vertices selected.
	vertical dimension / vertical (dimensional)	Applies a vertical length constraint restricted along the Y-axis.	One line or two vertices selected.
	angular constraint / angle (dimensional)	Applies an angular constraint in degrees between two lines.	Exactly two lines selected.
	radius constraint / radius	Applies a fixed radius constraint to an arc or circle.	One selected arc or circle.
	diameter constraint / diameter / radius	Applies a fixed diameter constraint to a circle.	One selected circle.
	coincident / coincidence	Fastens two vertices or points together at identical coordinates.	Two selected vertices or points.
	unified coincidence / coincidentunified	Applies automatic coincidence simultaneously across multiple points.	Two or more selected vertices.
	lock point / lock	Locks the absolute X and Y Cartesian coordinates of a sketch point.	One selected point or vertex.

Icon	Command	Description	Requirements
	point on object / pointonobject	Constrains a point or vertex to remain fixed on a line or curve.	One point and one line/curve selected.
	horizontal / horizontal (geometric)	Forces one or more lines to stay perfectly horizontal.	One or more selected lines.
	vertical / vertical (geometric)	Forces one or more lines to stay perfectly vertical.	One or more selected lines.
	auto hor/ver / horver	Automatically applies horizontal or vertical constraint based on closest slope.	Selected line in sketch.
	parallel	Forces two or more lines to maintain parallel alignment.	Two or more selected lines.
	perpendicular	Constrains two lines to form a strict 90° right angle.	Two selected lines.
	tangent	Creates a smooth tangential transition between a line and curve or two curves.	One line and arc/circle, or two arcs.
	equal	Equates lengths between lines or radii between arcs and circles.	Two or more selected lines or arcs.
	symmetric	Constrains two points to be symmetrical about a centerline or third point.	Two points and one symmetry line/axis.
	block / rigid block	Locks relative position and shape of an entity as a rigid body.	Selected geometry in sketch.
	toggle driving/reference / toggledriving	Toggles between driving constraint (defining size) and reference (read-only).	Selected dimensional constraint.
	toggle active constraints / toggleactive	Temporarily enables or disables selected constraints.	Selected constraints.

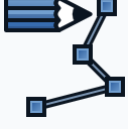
18. Draft Workbench – Tracing and Annotations

2D drawing entities, wires, ellipses, text, and dimensions.

Icon	Command	Description	Requirements
	wire / polyline / wire	Starts interactive creation of a 2D/3D polyline or Draft Wire.	Active working plane.
	map wire / createobjects	Maps and registers the active 2D wire into the accessible navigation system.	Newly created Draft Wire object.
	ellipse by center / center (ellipse)	Draws an ellipse defining center, major radius, and minor radius.	Active working plane.
	arc by center / center (arc)	Draws a circular arc from center point, radius, and angles.	Active working plane.
	arc by 3 points / points (arc)	Draws a circular arc defined sequentially across three points.	Active working plane.
	text	Inserts a 2D text annotation onto the active working plane.	Active document and coordinates.
	shape from text / shapestring	Creates a flat 2D geometric shape from text and a TrueType font (.ttf).	Active document and font file.
	label	Creates a 2D/3D descriptive label with leader line tied to an object or point.	Active document and selected point.
	style editor / editor	Opens configuration dialog for annotation fonts, sizes, and dimension styles.	None.
	linear dimension / linear	Inserts a linear measurement dimension between two points or along an edge.	Two points or one selected edge.
	flip dimension / flip	Inverts text orientation and arrow placement of the selected dimension.	Selected Draft dimension.
	facebinder	Creates a linked Facebinder object from selected surface geometries.	One or more selected faces.

19. Draft Workbench – Modification and Transformation

Geometric editing, cloning, projections, and advanced wire operations.

Icon	Command	Description	Requirements
	clone	Creates a linked parametric clone of the selected object.	Selected object.
	downgrade	Degrades or decomposes object to lower geometric level (solid to faces, etc.).	Selected object.
	draft to sketch / sketch	Converts Draft elements or polylines directly into a Sketcher sketch.	Selected 2D Draft object.
	edit / edit mode	Enters point and vertex editing mode for the selected element.	Selectable Draft object.
	fillet	Applies a fillet or rounding radius to the corners of a 2D object.	Selected polyline or corners.
	join	Joins multiple adjacent lines or edges into a single continuous wire.	Two or more connected edges.
	move	Translates selected objects from a base point to a target point.	Selected object.
	offset	Creates a parallel offset copy at a defined distance from original contour.	Selected wire or polyline.
	rotate	Rotates selected objects around a center point by a defined angle.	Selected object.
	mirror	Creates a symmetrical copy across a reflection plane or line.	Selected object.
	scale	Scales selected objects proportionally or along specific axes.	Selected object.
	shape 2D view / shape_2d_view	Generates a flat 2D projection from a 3D view or selected 3D object.	Selected 3D object.
	slope	Sets or alters the slope (Z-inclination) of a line or wire.	Selected line or wire.
	split	Splits a line or wire at a specified point into two separate entities.	Selected line/edge and point.
	stretch	Deforms and stretches geometry by displacing a set of vertices.	Selected object and vertices.
	highlight sub-elements / subelement_highlight	Toggles visual highlighting mode for faces, edges, or vertices.	Selected object.

Icon	Command	Description	Requirements
	trim / extend / trimex	Trims or extends an entity to the intersection with another geometry.	Selected line/wire.
	upgrade	Joins and promotes simple elements into closed faces or composite solids.	Selected geometries.
	wire to B-spline / wire_to_bspline	Converts a segmented polyline into a smooth B-Spline curve.	Selected wire/polyline.

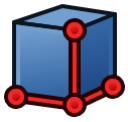
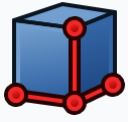
20. Draft Workbench – Point Placement

Precision placement commands for geometric reference points.

Icon	Command	Description	Requirements
	add point / addpoint	Places a reference point at the selected vertex or object's first vertex.	Object with vertices or selected vertex.
	point at vertex / pointatvertex	Inserts a point at the specific indicated vertex index.	Object with vertices and index number.
	midpoint	Calculates and places a point exactly at the midpoint of the selected edge.	Selected edge or object with edges.
	point at coordinates / pointatcoords	Creates a point at an exact spatial location via X, Y, Z coordinates.	Numerical X, Y, Z coordinate values.
	point placement help / help	Displays contextual help for point placement tools.	None.

21. Part Workbench (Solids, Primitives, and CSG Operations)

Parametric 3D primitives, machining operations, surfaces, sweeps, and boolean cuts.

Icon	Command	Description	Requirements
	box	Creates a parametric rectangular box/prism primitive (default 10x10x10 mm).	Active document.
	cube	Creates a solid parametric cube primitive (10x10x10 mm).	Active document.
	cylinder / primitive cylinder	Creates a solid cylinder specifying radius (2 mm) and height (10 mm).	Active document.

Icon	Command	Description	Requirements
	cone / primitive cone	Creates a solid cone with base radius (2 mm), top radius (4 mm), height (10 mm).	Active document.
	ellipse	Creates a parametric elliptical curve/plane with major radius (4 mm) and minor (2 mm).	Active document.
	circle	Creates a parametric circle primitive (radius 2 mm).	Active document.
	line	Creates a straight parametric line segment from (X1, Y1, Z1) to (X2, Y2, Z2).	Active document.
	new sketch / sketch	Inserts a new SketchObject into the active document.	Active document.
	chamfer	Applies a 1.0 mm beveled chamfer cut on the selected Part object.	Selected solid Part object.
	fillet	Applies a 1.0 mm fillet or blend radius to the selected solid.	Selected solid Part object.
	extrude / extrude object	Extrudes the selected Part object 10 mm along the Z-axis.	Selected Part object or 2D profile.
	color face / set face color / paint face	Opens color and transparency dialog to paint individual faces of the solid.	Selected Part object.
	cross sections	Opens dialog to generate cross-sectional cut curves across the object.	Selected Part object.
	loft / loft profiles	Generates a smooth lofted solid across two or more selected profiles.	Minimum 2 selected profiles/sections.
	make face / face	Creates a filled planar face from a selected closed wire or boundary.	Selected closed wire or boundary.
	mirror	Generates a mirrored copy of the selected object across the XY plane.	Selected Part object.

Icon	Command	Description	Requirements
	offset / 3d offset	Applies a 1.0 mm 3D thickening or spatial offset to the selected object.	Selected Part object.
	2D offset / offset 2d	Applies a 1.0 mm 2D planar offset to a selected wire or face.	Selected 2D wire or face.
	projection / project drawing	Projects one or more geometric shapes onto a support surface.	Support surface and at least one shape to project.
	revolve	Generates a solid of revolution revolving the object 360° around the Z-axis.	Selected Part object or profile.
	ruled surface	Creates a ruled surface bounded between two selected edges or curves.	Exactly 2 selected edges or curves.
	scale	Applies a uniform 2.0x scaling factor to the selected object.	Selected Part object.
	section / intersection	Calculates and creates the geometric intersection curve between two Part objects.	Exactly 2 selected Part objects.
	sweep / sweep profile	Generates a solid by sweeping the first profile along a guide trajectory.	Minimum 2 elements: profile and trajectory edge.

22. Voice Accessible Selection Tool (Selection)

Sequential object navigation and selection for accessibility.

Icon	Command	Description	Requirements
	next / forward / another	Advances to the next object in the list and highlights it on screen.	At least one object in the document.
	previous / back / return	Steps back to the previous object in the list and highlights it.	At least one object in the document.
	all / everything / select all	Simultaneously selects all objects present in the project.	Document containing objects.
	none / clear / deselect	Clears any active selection, leaving the scene unselected.	None.
	this / current / current object	Reports the name of the currently focused object via voice or console.	Initialized selection pointer.

Icon	Command	Description	Requirements
	count / total / quantity	Reports the total number of objects present in the document.	Active document.
	selection help / help	Displays contextual help for accessible selection commands.	None.



Guided examples

Learning by doing: the house, the bolt with its nut and the scissors

DAV ships with examples that are built step by step, by voice. Each frame tells you what to say; when you have said it all, in order, the action runs and the next frame appears. This section explains three of them.

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1. How to open an example

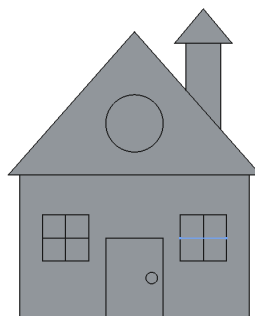
1. Say **explorer**, then **examples** and **examples** again: the picker opens.
2. With **next** (or **back**) go through the list to the example and say **send**.
3. A window, which does not block FreeCAD, shows one frame at a time with **what you would say to do it**: first the path through the menus and then the values. Each value is confirmed with **send**.
4. **skip** runs the frame without saying its words; **cancel** closes the example; at the end, **send** closes it.
5. Numbers from 0 to 99 are said naturally (“thirty two”); from 100 on, digit by digit. Decimals use “point”.

2. House: 2D shapes (Draft)

A house is drawn with 2D shapes only, with dictated sizes, and you learn to trim one shape with another.

Body	rectangle by corners, from (2, 0) to (42, 25)
Roof	triangle by vertices (0, 25), (44, 25) and (22, 50)
Door and windows	rectangle by corners from (17, 0) to (27, 14); two “rectangle by centre” of 8 × 8 at (10, 14) and (34, 14)
Chimney	rectangle from (31, 30) to (37, 48) and a triangle as its cap; trimmed with <i>modify</i> → <i>cut</i>
Details	round window (circle at (22, 34), radius 5), door knob (radius 1) and the window crosses with lines by points

Path: *workbench* → *draft* → *create* → *rectangle* / *triangle* / *circle*, and *up* → *drawing* → *line* for the lines.

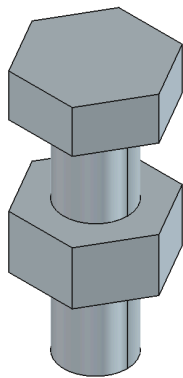


Result of the house example

3. Bolt and nut: an assembly (PartDesign and Assembly)

An M6 bolt (simplified from DIN 931) and its nut, made in PartDesign and joined in an assembly.

Shank	cylinder with radius 3 and height 20, centred at (0, 0, 10)
Head	6-sided prism, radius 6 and height 4, centred at (0, 0, 22), added to the same body
Nut	another body: 6-sided prism, radius 6 and height 5, at (15, 0, 4), with a subtractive cylinder of radius 3
Assembly	<i>create assembly. insert link twice, ground part (the bolt) and a cylindrical joint through the shank and hole cylinders; solve and the “isometric” view</i>



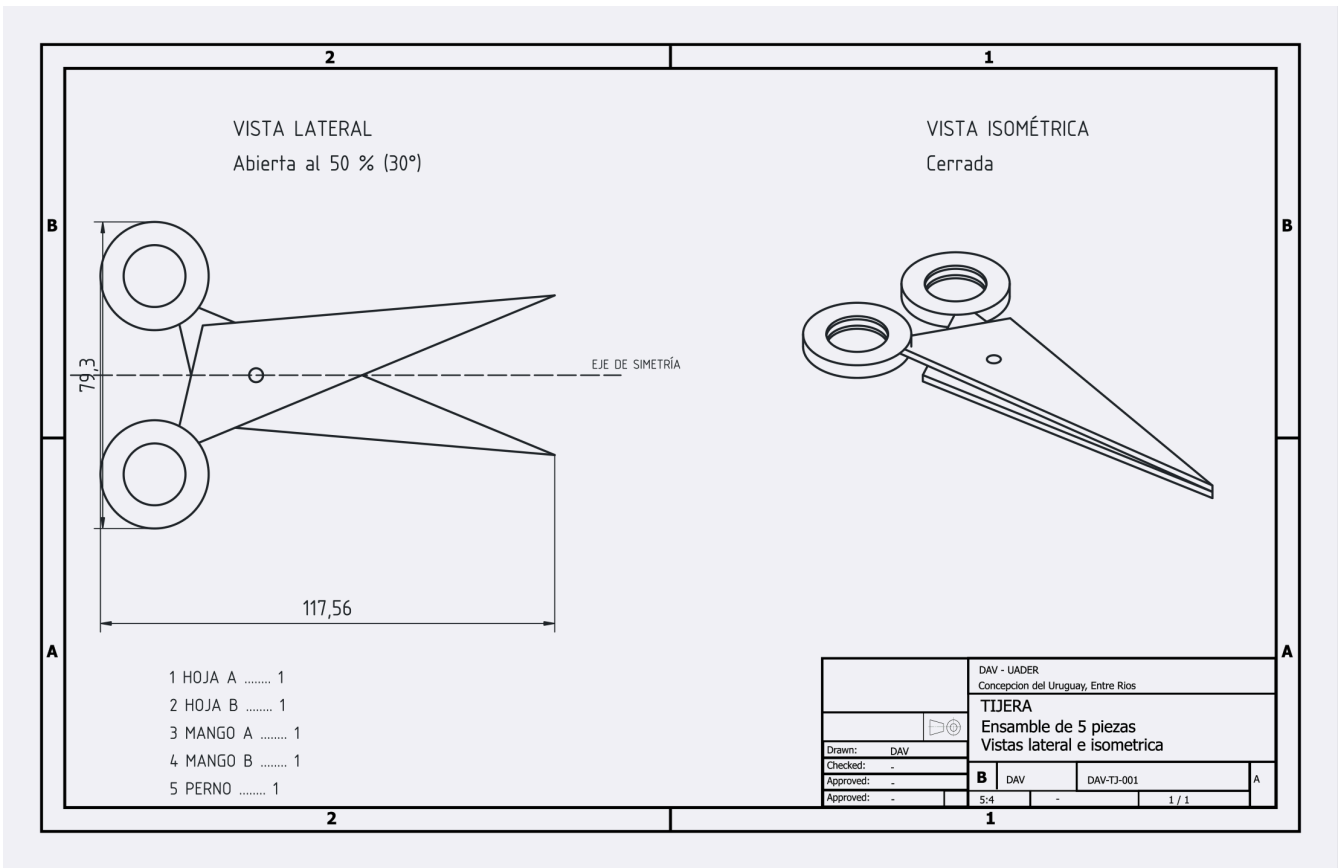
Bolt and nut, isometric view

4. Five-part scissors: assembly and drawing (PartDesign and Assembly)

Two blades, two handles and a pin: five parts in PartDesign, joined in an assembly with hinges and fixed joints.

Blades	triangle by vertices (80, 0), (-10, -16), (-26, 14) extruded 2 mm, plus a tang and two holes; blade B is its mirror in Y
Handles	rings of radius 14 and height 4, with a hollow of radius 8, at (-32, ±18)
Pin	stepped (radius 2 and 1.8): so the assembly stacks blade B 2 mm above A with no position joints
Assembly	5 × insert link, ground part (the pin), two hinge (blades) and two fixed joint (blade-handle); solve and “isometric”

The DAV example goes as far as the closed scissors. The guide *Dav/docs/guia-tijeras-voz.md* goes on with the scissors open at 50 % (a 23.3 mm distance joint between the handles) and with the technical drawing on an ANSI B sheet (ASME Y14.1, third-angle projection): side view with symmetry axis, isometric view and title block.



Scissors drawing: side view open at 50 % and closed isometric view

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